



Vision Mamba-enhanced Multi-level Context Aggregation Network for polyp segmentation

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ABSTRACT

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Colon adenomatous polyps represent a primary precursor to colorectal cancer (CRC), a disease with considerable global prevalence and mortality. Accurate segmentation of these polyps in colonoscopy images is therefore crucial for reliable detection and effective clinical intervention. Despite significant advancements, automated segmentation remains challenging due to substantial variations in polyp size, location, and morphology, as well as their often indistinct, blending boundaries with surrounding mucosa. To address these challenges, we propose VMMCA-Net, a novel Vision Mamba-enhanced Multi-level Context Aggregation Network. Our model is designed to enhance segmentation performance by integrating multi-level contextual information across scales. Specifically, we present a Hierarchical Adaptive Mamba Block (HAMB), which employs Vision Mamba to capture multi-scale representations and adapt to size variations in polyp regions. Additionally, a Cross-Feature Coupling Module (CFCM) is presented to fuse adjacent features across different network levels, thereby enriching contextual information. Furthermore, a Boundary-informed Conversion Module (BCM) is proposed to incorporate explicit edge-aware features into the architecture, leveraging boundary cues to enhance segmentation accuracy. Extensive experiments on five public datasets demonstrate that VMMCA-Net achieves state-of-the-art performance, outperforming existing methods in both quantitative metrics and qualitative visual assessments. The implementation code will be released at <https://github.com/jamescj98/VMMCA-Net>.

问题: ① 息肉在大小位置和形态上的显著差异
② 其与周围黏膜边界通常不明显, 容易混合

① 分层自适应 Mamba 块 HAMB

② 跨特征耦合模块 CFCM

③ 边界感知转换模块 BCM

1. Introduction

肿瘤学病理

结直肠腺癌

Among all oncological pathologies, colorectal adenocarcinoma is now recognized as the third most common cancer type by global incidence, with its development typically linked to adenomatous polyps [1]. If not addressed, these polyps can take 10 to 15 years to transition into cancerous lesions. Therefore, early detection and removal of polyps are crucial for preventing CRC and reducing mortality. Clinicians usually start the process by finding adenomatous polyps and then manually delineating them in colonoscopy images, which is a very difficult job that needs a lot of training. Unfortunately, this strategy takes a lot of time and can be subjective. This shows how important it is to have automated polyp segmentation algorithms that can help correctly segment polyp regions for later diagnostic purposes [2].

The wide range of size and form variation, as well as the diversity in features across developmental phases, makes the task of polyp segmentation extremely challenging [3,4]. Polyps and neighboring mucosal tissue are similar, making segmentation difficult. Recent advances in deep learning have yielded numerous innovative solutions for overcoming challenges, demonstrating considerable improvements in polyp segmentation [5]. In one notable approach, Akbari et al. [6] developed

a hybrid methodology combining fully convolutional networks with Otsu's thresholding algorithm, which particularly excels at isolating dominant polyp regions through connected component analysis. At the same time, Sun et al. [7] introduced a better FCN design that integrates dilated convolutional layers, successfully maintaining spatial details while acquiring comprehensive semantic representations via enlarged receptive fields. UNet [8] and its variants, distinguished by its encoder-decoder design, have produced notable breakthroughs. Improved semantic and contextual fusion is achieved in these models by progressively upsampling and merging high-level data from the decoder with corresponding low-level characteristics from the encoder via skip connections. Prominent variations, such as ResUNet++ [9] and UNet++ [10] are ideal for the purpose of polyp segmentation, demonstrating impressive performance. Nevertheless, several current methodologies focus on delineating the entire polyp area, thus overlooking essential border details. Certain research have attempted to address this problem by integrating boundary restrictions inside the region or employing multitask frameworks, resulting in enhanced segmentation accuracy [11]. Moreover, Fan et al. [12] used a backward

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attention module to make good use of border cues, which improved the accuracy of segmentation.

Although automated polyp segmentation has improved considerably, numerous challenges remain to be tackled. First, in scenarios involving flat lesions or suboptimal bowel preparation, the visual contrast between polyps and adjacent tissue may be insufficient, resulting in segmentation errors. Therefore, integrating boundary-aware information becomes crucial for accurately capturing the spatial relationship between polyp regions and their surrounding context. Second, scale variation poses a significant challenge, warranting further investigation into methodologies capable of effectively characterizing multi-scale information within convolutional layers. Lastly, although convolutional neural networks (CNNs) consist of multiple layers that operate at different scales, the shallower layers capture structural details. The deeper layers capture advanced semantic features essential for identifying polyp regions. Consequently, effectively integrating these deep semantic features with structural information remains a formidable challenge in producing accurate and reliable final segmentation maps.

While CNN-based approaches [13] excel at locality modeling, their fixed receptive fields fundamentally constrain their capacity to capture long-range dependencies, hindering performance on large-scale lesions or diffuse boundaries. Although Transformer-based methods [14, 15] address this through global self-attention, their quadratic complexity incurs prohibitive computational overhead, particularly on the high-resolution features typical of dense prediction. In contrast, Vision Mamba employs a selective state-space mechanism that achieves global context modeling with linear complexity. This efficiency makes it ideal for hierarchical aggregation under strict constraints, enabling our framework to maintain both precise boundaries and semantic coherence across scales.

In this paper, we propose a novel Vision Mamba-enhanced Multi-level Context Aggregation Network (VMMCA-Net) for polyp segmentation. Our framework uses multi-level and multi-scale features to gather semantic information and traits that span multiple scales. To effectively acquire and integrate multi-scale information inside the network, we provide a Hierarchical Adaptive Mamba Block (HAMB) that makes use of the Vision Mamba mechanism to capture long-range dependencies for boosting segmentation accuracy. To enhance cross-level feature interaction, we design the Cross-Feature Coupling Module (CFCM), which enables efficient integration of adjacent hierarchical features, facilitating robust scale-invariant context modeling. Moreover, the Boundary-informed Conversion Module (BCM) is developed to inject boundary-relevant cues into the segmentation process, refining spatial localization and structural delineation. Collectively, our framework synergistically combines boundary information with semantic features, strengthening feature hierarchies and improving segmentation fidelity. Experimental results on multiple colonoscopy datasets demonstrate that VMMCA-Net performs better than other state-of-the-art polyp segmentation methods. Furthermore, ablation studies confirm the individual contributions of each module, underscoring their importance in achieving the observed performance gains.

The main contributions of this paper are summarized as follows:

- ① 将边界线索与多层语义表示协同整合
 - We propose a novel Vision Mamba-enhanced Multi-level Context Aggregation Network (VMMCA-Net) that synergistically integrates boundary cues with multi-tier semantic representations to enable highly accurate polyp segmentation.
- ② 集成多尺度卷积和自适应注意力机制，捕捉边缘信息并有效整合多尺度特征表示
 - A Hierarchical Adaptive Mamba Block (HAMB) is presented to seamlessly integrate multi-scale convolutions and adaptive attention mechanisms, ensuring to capture long-range dependencies for enhancing feature representation.
- ③ 整合相邻层级特征，跨层融合
 - We develop a Cross-Feature Coupling Module (CFCM) that effectively integrates features from adjacent levels, facilitating robust cross-level fusion and addressing scale variations to improve segmentation accuracy.
- A Boundary-informed Conversion Module (BCM) is designed to capture and incorporate boundary contextual information into the segmentation process, thereby mitigating prediction inaccuracies.

④ 捕捉并将边界上下文信息纳入分割过程

2. Related works

In this section, we review three types of research works related to the proposed framework, consisting of medical image segmentation, segmentation of colonic polyps, and multi-scale/level feature representation.

2.1. Medical image segmentation

In computer-aided detection systems, medical image segmentation [16] is the most important step, as it allows for the identification of areas of interest and impacted regions. Recently, there have been substantial developments in CNN-based techniques originating from the broader field of pattern recognition, which have shown great potential in medical image segmentation [17,18]. There are a lot of different ways to divide biological images. The UNet architecture [8] stands out as a noteworthy model that has gained a significant amount of success. A great number of different variations of the UNet architecture have been proposed, each of which introduces upgrades with the intention of boosting the accuracy of segmentation. ResUNet++, for instance, was proposed by Jha et al. [9]. This new version of ResUNet improves upon the original ResUNet by incorporating more advanced components into the UNet framework. These components include squeeze-and-excitation modules and attention methods, which ultimately result in enhanced feature representation. In another effort, Zhou et al. [10] introduced UNet++, which refines the UNet design by employing redesigned skip connections to better align encoder and decoder features, effectively narrowing the semantic gap and addressing challenges associated with network depth.

2.2. Segmentation of colonic polyps

Historically, polyp segmentation approaches mostly relied on manually designed features, like color, shape, texture, and morphology, or combinations thereof [12,19]. Following the extraction of features, these methods often utilized different types of classifiers in order to differentiate polyps from the tissues that were surrounding them. However, the limited expressiveness of such manually engineered features often led to subpar segmentation outcomes. In a representative example, Tajbakhsh et al. [19] proposed an automated detection system that leveraged contextual and shape-related cues to suppress false positives and accurately localize polyps in colonoscopy videos. Despite these advances, the substantial variability in polyp texture and morphology across clinical scenarios continues to challenge traditional methods, restricting their effectiveness. The emergence of deep learning has led to the widespread adoption of fully convolutional networks (FCNs) for polyp segmentation tasks. For example, Sun et al. [7] created a model based on FCNs utilizing dilated convolutions to obtain high-level semantic information while maintaining spatial resolution. Furthermore, improved UNet versions like ResUNet++ [9] and UNet++ [10] have exhibited commendable efficacy in this field. Nonetheless, numerous algorithms concentrate on segmenting the entire polyp mass while sometimes overlooking detailed border information, which is essential for accurate localization. To mitigate this shortcoming, Fang et al. [11] proposed a boundary-aware loss function that enforces spatial limitations around polyp edges, hence improving segmentation accuracy.

2.3. Multi-scale/level feature representation

Multi-scale feature representations play an essential role in alleviating the issues posed by variations in object scale during detection and segmentation tasks [20,21]. Taking advantage of the natural correlations between visual patterns at different scales, Jiang et al. [22]

利用不同尺度下视觉模式之间的自然相关性

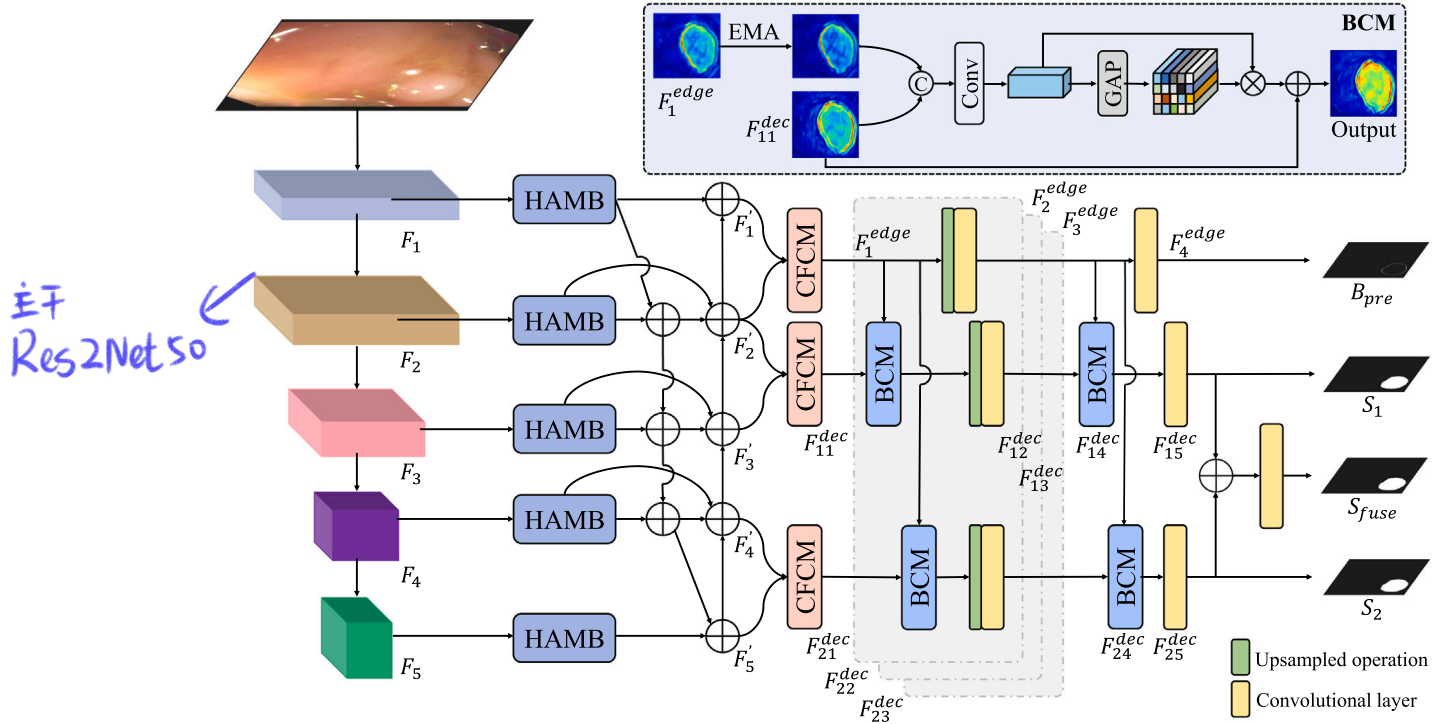


Fig. 1. Overview of the proposed VMMCA-Net framework, consisting of three key components: Hierarchical Adaptive Mamba Block, Cross-Feature Coupling Module, and Boundary-informed Conversion Module.

proposed a progressive fusion module to hierarchically integrate scale-specific features. Beyond single-task designs, multi-level fusion strategies have been widely explored across various computer vision domains. One representative line of work relies on shortcut-based aggregation, where feature maps from different stages are directly combined through skip connections, as seen in UNet [8] and UNet++ [10]. Although this strategy effectively bridges spatial and semantic information, it lacks discriminative selection and may introduce redundancy. To enable more adaptive cross-level interaction, Zhao et al. [23] employed spatial and channel recalibration mechanisms to guide hierarchical aggregation. Boundary-aware aggregation can also be viewed as a specialized form of multi-level feature fusion, in which high-frequency contour cues from shallow layers are combined with deep semantic features to enhance structural precision. Fan et al. [12] leveraged reverse attention to progressively refine ambiguous boundaries, while Yue et al. [24] enforced explicit contour supervision during fusion. In addition, Nguyen et al. [25] and Li et al. [26] further integrated coarse semantic priors with fine structural cues to improve cross-scale consistency. Complementary advances from natural image segmentation further inspired hierarchical fusion design. Chen et al. [27] aggregated multi-scale context using pooled or dilated features based on pyramid-based framework. Tan et al. [28] extended this concept through iterative top-down and bottom-up refinement, while Zheng et al. [29] explored long-range aggregation via transformer-based global self-attention, albeit with higher computational cost. Medical image analysis has equally benefited from both multi-scale modeling and multi-level feature fusion strategies. Sinha et al. [30] exploited multi-resolution encodings to strengthen contextual perception in lesion detection. Zhou et al. [31] further introduced a cross-level fusion framework that jointly leverages boundary cues and semantic hierarchies for accurate polyp segmentation. Unlike conventional CNN-based dual-stream fusion, our VMMCA-Net incorporates a Vision Mamba mechanism to enable bidirectional feature coupling with efficient long-range propagation. This design proves particularly advantageous for segmenting polyps with irregular structures or indistinct boundaries.

3. Methodology

3.1. Network overview

Fig. 1 shows the architecture of the proposed vision Mamba-enhanced multi-level context aggregation network for polyp segmentation, including three key components, i.e., the Hierarchical Adaptive Mamba Block (HAMB), Cross-Feature Coupling Module (CFCM), and Boundary-informed Conversion Module (BCM). These components can capture multi-scale, cross-domain, and boundary-specific information, resulting in the improved segmentation performance. Specifically, an input image $I \in \mathbb{R}^{W \times H \times 3}$ is initially fed into the encoder model (Res2Net-50 [32] as its backbone), to obtain multi-level features $\{F_i, i = 1, 2, \dots, 5\}$. Consequently, we obtain a feature resolution of $\frac{W}{4} \times \frac{H}{4}$ at the first level, and for every $i > 1$, we have a general resolution of $\frac{W}{2^i} \times \frac{H}{2^i}$. Then, the HAMB extracts multi-scale features using convolutions of varying sizes, enhancing spatial resolution and dependency capture. Moreover, the CFCM fuses cross-domain features via multi-scale convolutions and dilation, improving robustness and contextual understanding. Furthermore, the BCM refines features by combining contextual attention with boundary aggregation, enhancing discriminability and boundary accuracy. We will detail each key component below.

3.2. Hierarchical Adaptive Mamba Block

针对问题
A major obstacle in polyp segmentation lies in the severe variation of object scales, where polyps may appear with drastically different sizes and shapes across images. Conventional convolutional neural networks are inherently limited in this regard, as each convolutional layer operates at a fixed receptive field, restricting the ability to simultaneously model global contextual cues. To alleviate this issue, we propose the Hierarchical Adaptive Mamba Block to efficiently extract and integrate multi-scale spatial features while preserving long-range dependencies. This is achieved through a combination of convolutions with varying kernel sizes, channel partitioning, and residual learning strategies. As seen in Fig. 2, a 1×1 convolutional layer is first

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applied to F_i to reduce its channel dimension and alleviate computational overhead. The resulting feature map is then evenly partitioned along the channel dimension into four branch features, denoted as $\{X_1, X_2, X_3, X_4\}$. Each branch feature X_i is independently processed by a Mamba block to capture long-range spatial dependencies and contextual information. The Mamba block [33] consists of two linear layers, a Spatial Selective Mechanism (SSM) module for dynamic state updating, a SiLU activation function, and a 1×1 convolution for channel mixing, producing an enhanced feature representation for each branch. To preserve original spatial details and stabilize feature updates, each branch introduces a learnable modulation parameter r_i , which adaptively scales the original feature X_i via element-wise multiplication. This mechanism regulates the relative contributions of the original and Mamba-enhanced features, enabling a harmonious integration of local and global information. The modulated features are further refined using convolutional layers with progressively increasing kernel sizes $b_i \in \{3, 5, 7, 9\}$. Specifically, the above processing steps can be summarized as follows:

$$\begin{cases} Y_i = \mathcal{B}_{Conv b_i \times b_i}(\text{Mamba}(F_i) \oplus (X_i \otimes r_i)), & i = 1, \\ Y_i = \mathcal{B}_{Conv b_i \times b_i}(\text{Mamba}(F_i) \oplus (X_i \otimes r_i) \oplus Y_{i-1}), & i = 2, 3, 4, \end{cases} \quad (1)$$

where $\mathcal{B}_{Conv b_i \times b_i}(\cdot)$ denotes a batch normalization and activation function followed by a $b_i \times b_i$ convolution, $\oplus$ represents element-wise addition, and $\otimes$ denotes element-wise multiplication. The hierarchical dependency across branches enables larger kernels to refine intermediate features from smaller receptive fields, facilitating efficient multi-scale feature integration. The outputs from all branches $\{Y_1, Y_2, Y_3, Y_4\}$ are concatenated along the channel dimension and fused via a 1×1 convolution. Finally, a residual connection is applied to combine the fused features with the original input, yielding the final output of the HAMB:

$$F_i^{\text{HAMB}} = \mathcal{B}_{Conv 1 \times 1}(F_i \oplus \text{CAT}(Y_1, Y_2, Y_3, Y_4)). \quad (2)$$

The progressive use of enlarged convolutional kernels in HAMB is designed to balance representational diversity and computational efficiency. Smaller kernels capture fine-grained boundary details crucial for irregular polyp delineation, while larger kernels aggregate broader contextual information to stabilize predictions in low-contrast regions or for large lesions, thereby avoiding the bias toward either local sharpness or global smoothness inherent in single-kernel designs.

Unlike existing Mamba-based vision methods [33,34], which typically model long-range dependencies at a fixed spatial scale, HAMB embeds Mamba within a hierarchical multi-scale architecture. Features from different receptive fields are progressively refined and integrated, allowing larger kernels to enhance intermediate representations from smaller scales instead of recomputing full-resolution features. Combined with residual learning, this design enables efficient joint modeling of long-range dependencies and scale-aware spatial variations, ensuring robust feature learning and stable optimization in complex segmentation scenarios.

3.3. Cross-feature coupling module

Fig. 3 presents the structure of the proposed CFCM. This module aims to improve the extraction and integration of cross-domain feature information by combining multi-scale convolutions with multi-dilation encoding layers. Conventional cross-level fusion methods (e.g., addition or concatenation) simply stack features and treat all channels equally, without considering semantic inconsistency between adjacent decoder stages. In medical images, low-level textures and high-level semantics often exhibit domain discrepancy, making naive fusion insufficient for reliable feature coupling. To address this issue, CFCM is designed as a content-aware interaction module in contrast to a static fusion operator. By decomposing inputs into multi-dilation subspaces and performing learnable cross-correlation matching, it adaptively strengthens compatible responses while suppressing scale-inconsistent noise. This

enables CFCM to negotiate cross-level dependencies more effectively than conventional fusion schemes. The CFCM integrates two different input features and employs an effective learning strategy to emphasize relevant feature interactions. We project two input feature maps F_i and F_{i+1} to lower-dimensional spaces, denoted F'_i and F'_{i+1} , using a 1×1 convolutional layer $\mathcal{B}_{Conv 1 \times 1}(\cdot)$. This operation reduces the computational load from redundant channels and ensures consistency in channel dimensions for subsequent multi-branch operations. We use two separate but complementary approaches to gather contextual data at varying sizes, i.e., $\text{Enc3}(\cdot)$ and $\text{Enc5}(\cdot)$. These branches employ 3×3 and 5×5 convolutional kernels, respectively, in conjunction with multiple dilation rates (e.g., $(1, 1)$, $(2, 1)$, $(1, 2)$). Therefore, each branch encodes both local and larger receptive fields along the spatial dimensions. In this manner, each input stream is transformed into multiple sub-features, covering various scales and aspect ratios. As a result, we can obtain

$$[z_{11}, z_{12}, z_{21}] = \text{Enc3}(\mathcal{B}_{Conv 3 \times 3}(F'_i)), \quad [x_{11}, x_{12}, x_{21}] = \text{Enc3}(\mathcal{B}_{Conv 5 \times 5}(F'_{i+1})), \quad (3)$$

$$[z'_{11}, z'_{12}, z'_{21}] = \text{Enc5}(\mathcal{B}_{Conv 3 \times 3}(F'_i)), \quad [x'_{11}, x'_{12}, x'_{21}] = \text{Enc5}(\mathcal{B}_{Conv 5 \times 5}(F'_{i+1})). \quad (4)$$

To enhance the matching of features at a fine-grained level, the GroupDWC($\cdot$) is used to compute pairwise *depthwise cross-correlation* between these sub-features. Specifically, we use grouped convolutions (groups = $B \times C$) to calculate the cross-correlation for each channel. Since there are three pairs of sub-features in each branch (e.g., x_{11}, x_{12}, x_{21} vs. z_{11}, z_{12}, z_{21}), we design a learnable weight vector $\mathbf{w} = [w_1, w_2, w_3]^T$. After applying the softmax function, $\alpha_i = \text{Softmax}(w_i)$, the final fused feature of one branch can be obtained by

$$S = \sum_{i=1}^3 \alpha_i \cdot \text{GroupDWC}(\mathbf{x}_i, \mathbf{z}_i). \quad (5)$$

This learnable fusion automatically balances different dilation sub-paths, combining the advantages of local and larger receptive fields. After obtaining the two fused features (i.e., S_1 and S_2), we concatenate them (i.e., S_{cat}) and feed them into two different convolutional layers to further integrate the information. Subsequently, we apply elementwise multiplication and a skip connection from the aligned input features. Therefore, we can obtain the output of the i th CFCM as follows:

$$F_i^{\text{CFCM}} = \mathcal{B}_{Conv 3 \times 3}(\mathcal{B}_{Conv 3 \times 3}(F'_i) \oplus \mathcal{B}_{Conv 5 \times 5}(F'_{i+1}) \oplus \mathcal{B}_{Conv 3 \times 3}(S_{cat}) \otimes \mathcal{B}_{Conv 5 \times 5}(S_{cat})). \quad (6)$$

It can be noted that this residual-like structure preserves the primary information from both inputs while enhancing key cross-feature interactions in the fused domain. Consequently, the CFCM encapsulates both local and scale-dependent correlations, providing a discriminative feature representation for subsequent network stages.

3.4. Boundary-informed conversion module

It is essential to maximize the utilization of boundary-enhanced features while minimizing the redundancy and noise inherent in boundary-aware features. Effectively integrating these features into the segmentation decoder is crucial for improved performance. Previous methods such as PraNet [12] and BUNet [24] typically utilize boundary predictions as auxiliary attention masks or impose auxiliary losses during training. In contrast, we treat boundary cues as mere supervision signals. Instead, it converts the boundary response into a modulation prior that directly interacts with semantic activations, enabling feature-level coupling rather than pixel-wise masking. This conversion mechanism allows boundaries to actively reshape semantic representations, leading to more coherent region-edge consistency. Therefore, we present a Boundary-informed Conversion Module (BCM) to extract valuable

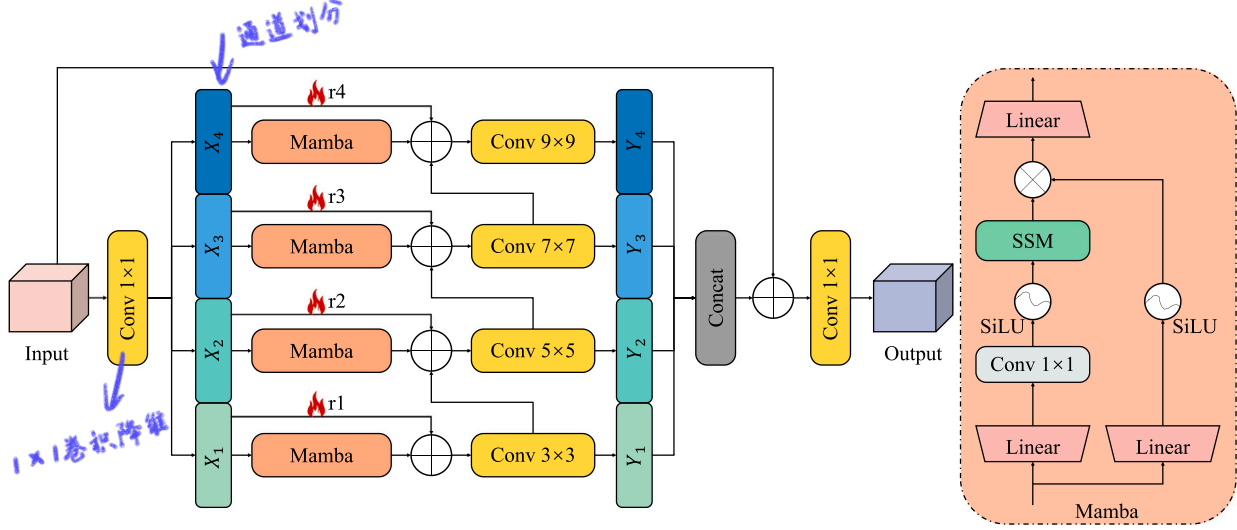


Fig. 2. Structure of the proposed Hierarchical Adaptive Mamba Block.

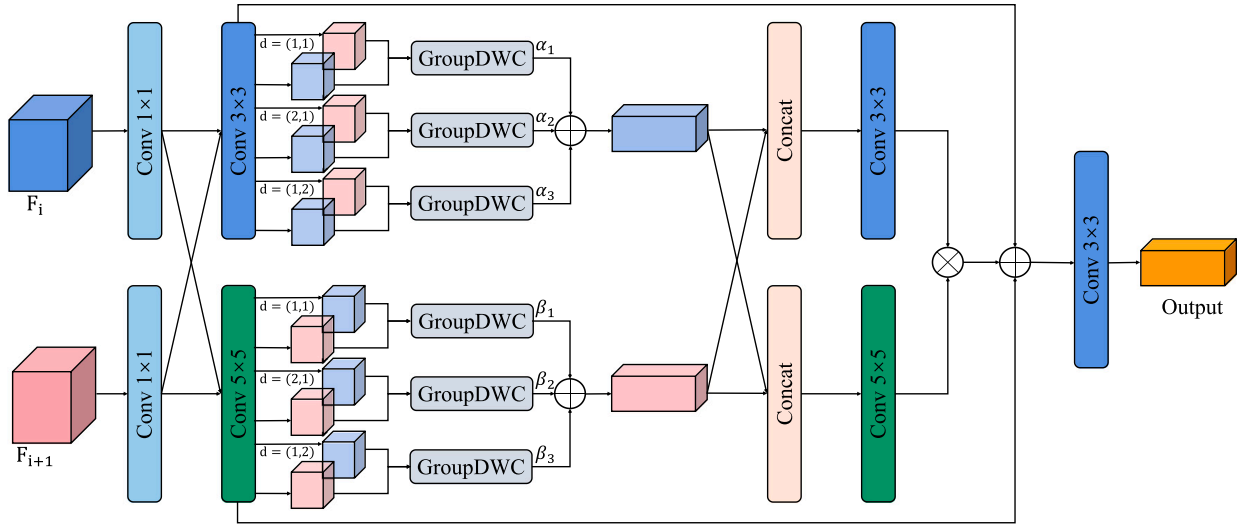


Fig. 3. Architecture of the proposed Cross-Feature Coupling Module.

information from boundary-aware features and seamlessly integrate it into the network. The procedure starts with a boundary-aware feature that is guided by the Efficient Multi-scale Attention (EMA) block [35], as shown in Fig. 2. This block refines the feature maps by employing multi-scale pooling and attention mechanisms, capturing long-range dependencies and enhancing contextual information. To supervise the boundary-aware branch, we derive binary boundary maps from the ground-truth segmentation masks. Specifically, we follow the strategy in [36] and first extract the object contours using morphological gradient operations. Then, an inward erosion process with a fixed width of 8 pixels is applied to generate boundaries of uniform thickness, ensuring that they remain within the valid mask regions. These maps serve as the supervision signals for the boundary stream during training. Consequently, this refined feature processing is outlined as follows:

$$F_{1,EMA}^{\text{edge}} = \text{EMA} \left(F_1^{\text{edge}} \right) \otimes F_1^{\text{edge}}. \quad (7)$$

Subsequently, the attention-refined boundary-aware feature is fused with the corresponding decoder output via a channel-wise concatenation operation, formulated as $F_{11}^{\text{cat}} = \text{CAT} \left(F_{1,EMA}^{\text{edge}}, F_{11}^{\text{dec}} \right)$. After being refined locally by a 3×3 convolution, the feature map that comes out of it is then put through a global average pooling (GAP) process. Readjusting the combined features by use of element-wise multiplication and a residual

link to the decoder output is accomplished by means of the GAP result acting as a modulation signal. We may mathematically represent this procedure as follows:

$$F_{11}^{BCM} = \mathcal{P}_{\text{ave}} \left(F_{11}^{\text{cat}} \right) \otimes F_{11}^{\text{cat}} \oplus F_{11}^{\text{dec}}, \quad (8)$$

where $\mathcal{P}_{\text{ave}}$ denotes the GAP operation and F_{11}^{BCM} represents the enhanced feature after Boundary-informed Conversion Module (BCM). This output is subsequently upsampled and propagated forward. Notably, the integration of boundary cues into the decoder pathway effectively augments the representation of structural details, contributing to improved segmentation accuracy.

The proposed polyp segmentation framework incorporates dual decoders, each supervised independently using the ground truth annotations. The last segmentation prediction is made by feeding the terminal outputs of the two decoders, F_{15}^{dec} and F_{25}^{dec} , into a $\mathcal{B}_{\text{Conv}3 \times 3}(\cdot)$ convolutional block. This operation aggregates multi-scale contextual information and yields the final prediction map.

3.5. Loss function

To achieve both detailed pixel-level precision and comprehensive structural consistency in polyp segmentation, we propose a meticulously designed loss function. Although the binary cross-entropy (BCE)

loss is widely used in segmentation tasks due to its simplicity, it inherently treats each pixel in isolation, thereby neglecting broader contextual relationships. To overcome this limitation, the segmentation loss is defined as:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{wIoU}} + \mathcal{L}_{\text{wBCE}}, \quad (9)$$

where $\mathcal{L}_{\text{wIoU}}$ and $\mathcal{L}_{\text{wBCE}}$ represent the weighted Intersection over Union (wIoU) and weighted Binary Cross-Entropy (wBCE) losses, which capture global and local constraints, respectively.

The $\mathcal{L}_{\text{wIoU}}$ enhances the importance of hard-to-segment regions by assigning higher weights to challenging pixels, and is formulated as:

$$\mathcal{L}_{\text{wIoU}} = 1 - \frac{\sum_{i=1}^W \sum_{j=1}^H (g_{i,j} \cdot p_{i,j}) \cdot (1 + 5\mu_{i,j})}{\sum_{i=1}^W \sum_{j=1}^H (g_{i,j} + p_{i,j} - g_{i,j} \cdot p_{i,j}) \cdot (1 + 5\mu_{i,j})}, \quad (10)$$

where $g_{i,j}$ and $p_{i,j}$ denote the values at pixel (i, j) in the ground truth and predicted segmentation maps, respectively, and $\mu_{i,j}$ represents a contextual weighting factor that emphasizes boundary and difficult regions. This formulation ensures that pixels near edges or with high uncertainty contribute more significantly to the overall loss, leading to improved global segmentation accuracy.

The $\mathcal{L}_{\text{wBCE}}$ prioritizes hard-to-classify pixels rather than treating all pixels equally, and is defined as:

$$\mathcal{L}_{\text{wBCE}} = - \frac{\sum_{i=1}^W \sum_{j=1}^H (1 + 5\mu_{i,j}) \sum_{l=0}^1 \mathbf{1}(g_{i,j} = l) \log P(p_{i,j} = l | \theta)}{\sum_{i=1}^W \sum_{j=1}^H (1 + 5\mu_{i,j})}, \quad (11)$$

where $\mathbf{1}(\cdot)$ is the indicator function, $l \in \{0, 1\}$ represents binary labels, and $P(p_{i,j} = l | \theta)$ denotes the predicted probability given model parameters θ . By assigning greater importance to difficult pixels, $\mathcal{L}_{\text{wBCE}}$ effectively improves local segmentation accuracy and stability.

To further refine prediction boundaries and maintain structural coherence, we introduce a weighted Dice loss $\mathcal{L}_{\text{wDice}}$, which emphasizes overlap between the predicted and ground truth boundary regions while weighting boundary pixels more heavily:

$$\mathcal{L}_{\text{wDice}} = 1 - \frac{2 \sum_{i=1}^W \sum_{j=1}^H g_{i,j} \cdot p_{i,j} \cdot (1 + 5\mu_{i,j})}{\sum_{i=1}^W \sum_{j=1}^H (g_{i,j}^2 + p_{i,j}^2) \cdot (1 + 5\mu_{i,j})}. \quad (12)$$

This formulation strengthens the precision of the boundaries by penalizing misclassified pixels near the edges, thereby improving contour delineation and structural fidelity. The proposed boundary loss $\mathcal{L}_{\text{boundary}}$ is defined as the combination of the weighted BCE and weighted Dice losses:

$$\mathcal{L}_{\text{boundary}} = \mathcal{L}_{\text{wBCE}} + \mathcal{L}_{\text{wDice}}. \quad (13)$$

Moreover, to align with the ground truth segmentation map G_s , three segmentation predictions (S_1^{up} , S_2^{up} , and $S_{\text{fuse}}^{\text{up}}$) are up-sampled to match their corresponding resolutions. The overall optimization objective is then formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{boundary}}(B_{\text{pre}}, G_b) + \sum_{i=1}^2 \mathcal{L}_{\text{seg}}(S_i^{\text{up}}, G_s) + \mathcal{L}_{\text{seg}}(S_{\text{fuse}}^{\text{up}}, G_s), \quad (14)$$

where $\mathcal{L}_{\text{boundary}}$ emphasizes the accuracy of boundary localization, while $\mathcal{L}_{\text{seg}}$ enforces consistency at both local and global scales. G_b is the ground truth boundary map, B_{pre} is the predicted edge. This comprehensive loss formulation allows the network to effectively capture fine-grained boundary details while maintaining structural completeness, thereby leading to superior segmentation performance.

4. Experiments

4.1. Datasets and evaluation metrics

Datasets. We conduct comparison experiments on five benchmark datasets to evaluate our method's effectiveness, and the details of each dataset are provided below. CVC-ClinicDB [37]: This dataset consists

Table 1

Six evaluation measures are used to look at the quantitative results on the CVC-ClinicDB dataset [37].

Methods	$mDice \uparrow$	$mIoU \uparrow$	$F_{\beta}^w \uparrow$	$S_a \uparrow$	$E_{\phi}^{\max} \uparrow$	$\mathcal{M} \downarrow$
UNet [8]	0.823	0.755	0.811	0.889	0.954	0.019
SFA [11]	0.700	0.607	0.647	0.793	0.885	0.042
UNet++ [10]	0.794	0.729	0.785	0.873	0.931	0.022
ACSNet [42]	0.889	0.826	0.873	0.927	0.959	0.007
PraNet [12]	0.899	0.849	0.896	0.936	0.969	0.006
MSNet [23]	0.918	0.869	0.913	0.946	0.974	0.008
C2FNet [43]	0.919	0.872	0.913	0.946	0.976	0.008
CCBANet [25]	0.868	0.812	0.861	0.916	0.944	0.020
LDNet [44]	0.881	0.825	0.879	0.924	0.965	0.012
CFA-Net [31]	0.933	0.883	0.924	0.950	0.989	0.007
PFENet [45]	0.922	0.876	—	—	—	—
NPD-Net [46]	0.921	0.872	0.914	0.943	0.967	0.008
PRCNet [26]	0.925	0.869	0.916	0.943	0.980	0.009
BUNet [24]	0.919	0.870	—	—	—	0.011
VMMCA-Net (Ours)	0.936	0.892	0.930	0.949	0.982	0.007

of 612 frames, each with a resolution of 288×384 pixels. ETIS [38]: It contains 196 polyp images, each with dimensions of 966×1225 pixels. CVC-ColonDB [19]: This dataset includes 380 images of polyp scenes, and each image is 500×570 pixels in size. Kvasir [39]: This dataset features 1000 images collected from various colonoscopy sequences and exhibits a diverse range of polyp appearances. CVC-300 [40]: It consists of 60 polyp images, each with a resolution of 500×574 pixel format.

Evaluation Metrics. In this study, we adopt two popular metrics for medical image segmentation, namely IoU and Dice. Additionally, four metrics are introduced, including S-measure (S_a), the F-measure (F_{β}^w), the E_{ϕ} , and the mean absolute error ($\mathcal{M}$), which are widely used in the field of object detection [41]. These metrics provide a comprehensive evaluation framework for assessing segmentation performance in both medical image segmentation and object detection.

4.2. Implementation details

Our architecture is built using PyTorch and was trained on a single 48 GB NVIDIA Tesla A40 GPU. We use a pretrained Res2Net-50 [32] as backbone. The model is fine-tuned with the help of the AdamW algorithm. The learning rate starts with an initial value of $5e-5$. The input images are downsized to 352×352 pixels, and then data augmentation methods such as random flipping, cropping, and rotation are applied to them. This model has been trained using a range of scaling ratios, namely $\{0.75, 1, 1.25\}$. With a batch size of 16, the model undergoes training for more than 300 epochs. The training dataset includes 550 images from CVC-ClinicDB and 900 images from Kvasir, following the experimental setup in [12].

4.3. Comparison with state-of-the-art methods

4.3.1. Comparison methods

In order to assess how well the suggested polyp segmentation approach worked, we compared it to current best practices, namely UNet [8], UNet++ [10], SFA [11], PraNet [12], ACSNet [42], MSNet [23], C2FNet [43], CCBANet [25], LDNet [44], CFA-Net [31], PFENet [45], NPD-Net [46], PRCNet [26] and BUNet [24]. To ensure a fair comparison, all compared approaches were reproduced using their official implementations or reported configurations with the appropriate performing hyperparameters recommended by the original authors.

4.3.2. Quantitative comparison

In Tables 1 and 2, we compare our suggested model to the best existing methods using six different performance measures and the CVC-ClinicDB [37] and Kvasir [39] datasets. The test results on the CVC-ClinicDB dataset show that our method works much better than

Table 2

Six evaluation measures are used to look at the quantitative results on the Kvasir [39] dataset.

Methods	$mDice \uparrow$	$mIoU \uparrow$	$F_{\beta}^w \uparrow$	$S_a \uparrow$	$E_{\phi}^{\max} \uparrow$	$\mathcal{M} \downarrow$
UNet [8]	0.818	0.746	0.794	0.858	0.893	0.055
SFA [11]	0.723	0.611	0.670	0.782	0.849	0.075
UNet++ [10]	0.821	0.744	0.808	0.862	0.910	0.048
ACSNet [42]	0.898	0.838	0.882	0.920	0.952	0.032
PraNet [12]	0.898	0.840	0.885	0.915	0.948	0.030
MSNet [23]	0.905	0.849	0.892	0.923	0.954	0.028
C2FNet [43]	0.886	0.831	0.870	0.905	0.935	0.036
CCBANet [25]	0.853	0.777	0.831	0.887	0.918	0.047
LDNet [44]	0.887	0.821	0.869	0.905	0.945	0.031
CFA-Net [31]	0.915	0.861	0.903	0.924	0.962	0.023
PFENet [45]	0.910	0.858	–	–	–	–
NPD-Net [46]	0.900	0.844	0.887	0.917	0.945	0.026
PRCNet [26]	0.799	0.716	0.763	0.837	0.886	0.068
BUNet [24]	0.898	0.841	–	–	–	0.026
VMMCA-Net (Ours)	0.922	0.874	0.910	0.929	0.963	0.021

Table 3

Six evaluation measures are used to look at the quantitative results on the CVC-300 [40] dataset.

Methods	$mDice \uparrow$	$mIoU \uparrow$	$F_{\beta}^w \uparrow$	$S_a \uparrow$	$E_{\phi}^{\max} \uparrow$	$\mathcal{M} \downarrow$
UNet [8]	0.710	0.627	0.684	0.843	0.876	0.022
SFA [11]	0.467	0.329	0.341	0.640	0.817	0.065
UNet++ [10]	0.707	0.624	0.687	0.839	0.898	0.018
ACSNet [42]	0.863	0.787	0.825	0.923	0.968	0.013
PraNet [12]	0.871	0.797	0.843	0.925	0.972	0.010
MSNet [23]	0.865	0.799	0.848	0.926	0.953	0.010
C2FNet [43]	0.874	0.801	0.844	0.927	0.968	0.009
CCBANet [25]	0.888	0.815	0.862	0.935	0.980	0.008
LDNet [44]	0.869	0.793	0.841	0.923	0.965	0.009
CFA-Net [31]	0.893	0.827	0.875	0.938	0.978	0.008
PFENet [45]	0.901	0.839	–	–	–	–
NPD-Net [46]	0.900	0.835	0.887	0.938	0.973	0.006
PRCNet [26]	0.819	0.736	0.777	0.891	0.909	0.018
BUNet [24]	0.891	0.820	–	–	–	0.006
VMMCA-Net (Ours)	0.902	0.841	0.882	0.941	0.966	0.007

Table 4

Six evaluation measures are used to look at the quantitative results on the CVC-ColonDB [19] dataset.

Methods	$mDice \uparrow$	$mIoU \uparrow$	$F_{\beta}^w \uparrow$	$S_a \uparrow$	$E_{\phi}^{\max} \uparrow$	$\mathcal{M} \downarrow$
UNet [8]	0.504	0.436	0.491	0.710	0.781	0.059
SFA [11]	0.456	0.337	0.366	0.629	0.754	0.094
UNet++ [10]	0.482	0.408	0.467	0.693	0.764	0.061
ACSNet [42]	0.716	0.649	0.697	0.829	0.851	0.039
PraNet [12]	0.712	0.640	0.699	0.820	0.872	0.043
MSNet [23]	0.751	0.671	0.736	0.838	0.883	0.041
C2FNet [43]	0.724	0.650	0.705	0.826	0.868	0.044
CCBANet [25]	0.706	0.626	0.676	0.812	0.863	0.052
LDNet [44]	0.740	0.652	0.717	0.830	0.884	0.036
CFA-Net [31]	0.743	0.665	0.728	0.835	0.898	0.039
PFENet [45]	0.742	0.668	–	–	–	–
NPD-Net [46]	0.701	0.630	0.689	0.815	0.848	0.040
PRCNet [26]	0.688	0.599	0.660	0.800	0.843	0.051
BUNet [24]	0.736	0.664	–	–	–	0.038
VMMCA-Net (Ours)	0.766	0.694	0.743	0.847	0.879	0.034

all the other comparison methods. CFANet and PRCNet are better at segmentation than other methods that are used to compare them. Additionally, SFA and BUNet take into account the interdependence between regions and edges; however, they remain ineffective in segmenting polyps, whereas our model successfully segments them and achieves superior performance. Our proposed model efficiently captures multi-scale features to tackle the scale variation of polyps, while incorporating boundary-aware information to improve segmentation

Table 5

Six evaluation measures are used to look at the quantitative results on the ETIS-LaribPolypDB [38] dataset.

Methods	ETIS-LaribPolypDB					
	$mDice \uparrow$	$mIoU \uparrow$	$F_{\beta}^w \uparrow$	$S_a \uparrow$	$E_{\phi}^{\max} \uparrow$	$\mathcal{M} \downarrow$
UNet [8]	0.398	0.335	0.366	0.684	0.740	0.036
SFA [11]	0.297	0.217	0.231	0.557	0.633	0.109
UNet++ [10]	0.401	0.344	0.390	0.683	0.776	0.035
ACSNet [42]	0.578	0.509	0.530	0.754	0.764	0.059
PraNet [12]	0.628	0.567	0.600	0.794	0.841	0.031
MSNet [23]	0.723	0.652	0.677	0.845	0.890	0.020
C2FNet [43]	0.699	0.624	0.668	0.827	0.875	0.022
CCBANet [25]	0.559	0.483	0.513	0.751	0.800	0.036
LDNet [44]	0.645	0.551	0.600	0.788	0.847	0.023
CFA-Net [31]	0.732	0.655	0.693	0.845	0.892	0.014
PFENet [45]	0.715	0.648	–	–	–	–
NPD-Net [46]	0.628	0.565	0.604	0.790	0.823	0.017
PRCNet [26]	0.685	0.598	0.639	0.824	0.851	0.022
BUNet [24]	0.692	0.614	–	–	–	0.016
VMMCA-Net (Ours)	0.752	0.679	0.708	0.846	0.894	0.016

Table 6

Quantitative results on the Kvasir, CVC-300, and CVC-ColonDB datasets, compared against Vision Mamba-based methods, using two evaluation metrics.

Methods	Kvasir		CVC-300		CVC-ColonDB	
	$mDice \uparrow$	$mIoU \uparrow$	$mDice \uparrow$	$mIoU \uparrow$	$mDice \uparrow$	$mIoU \uparrow$
VM-UNet [34]	0.879	0.783	0.841	0.726	0.694	0.531
VM-UNetV2 [47]	0.881	0.788	0.866	0.763	0.700	0.539
RM-UNet [48]	0.903	0.822	0.884	0.792	0.758	0.654
VMMCA-Net (Ours)	0.922	0.874	0.902	0.841	0.766	0.694

accuracy by offering boundary cues. Our polyp segmentation algorithm consistently outperforms existing methods on the Kvasir dataset. Specifically, on the two most important indicators, mDice and mIoU, our model demonstrates improvements of 2.4% and 3.3% over BUNet, respectively.

As shown in Tables 3 and 4, it can be observed that our model performed better than the others on the CVC-300 dataset. Although PFENet and NPD-Net demonstrate superior segmentation performance compared to other comparative methods, NPD-Net, despite expanding the scope of context information acquisition, remains ineffective in segmenting polyps. In contrast, our model successfully segments polyps by delivering boundary-aware features that facilitate the identification of polyp boundaries, resulting in precise segmentation and enhanced performance. On the CVC-ColonDB dataset, our model outperforms all other approaches by 3% on the mDice and mIoU metrics, which are the two most important indicators, compared to BUNet.

Table 5 shows a comparison of our model on the ETIS [38] dataset with six top-tier techniques. By obtaining improvements of 2% in mDice and 2.4% in mIoU over CFA-Net, our model is more adept at detecting intricate polyp edges. Simultaneously, VMMCA-Net introduces Vision Mamba for effectively capturing long-range dependencies, crucial for segmenting polyps with unclear boundaries or dispersed structures. VMMCA-Net is not just an iteration, but a significant leap forward in medical image segmentation through state-of-the-art state-space modeling.

4.3.3. Qualitative comparison

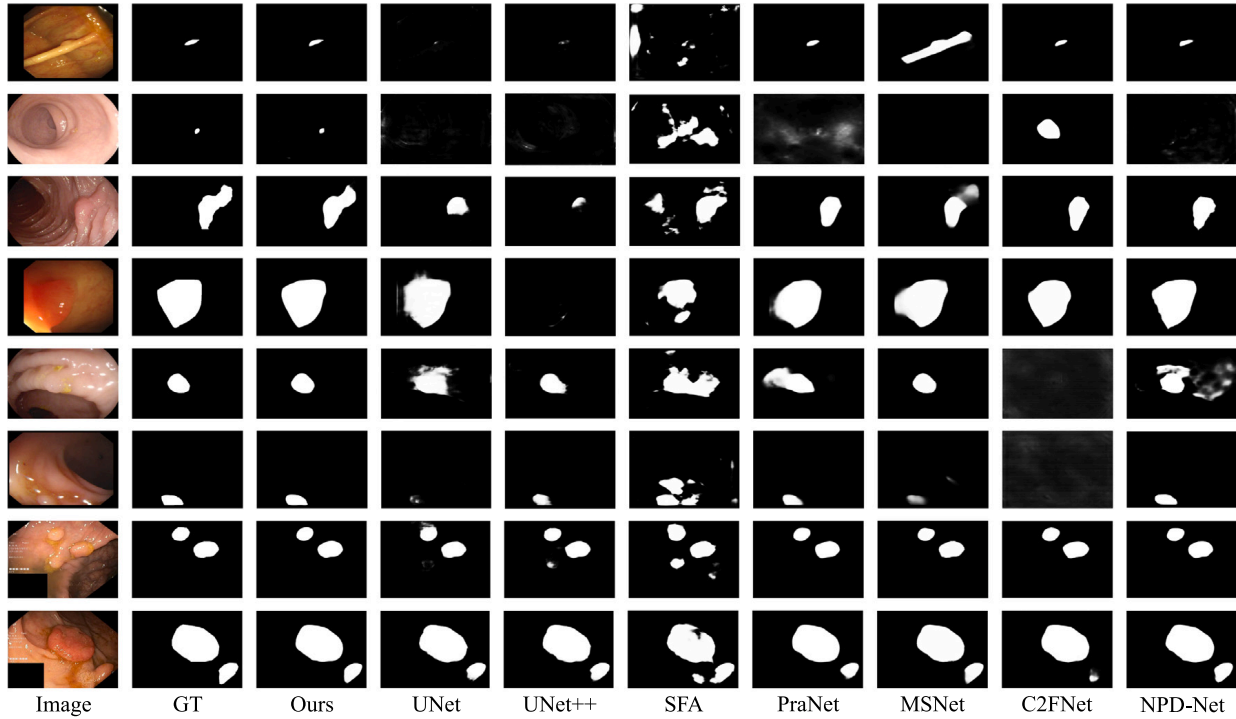
By contrasting our approach with cutting-edge polyp segmentation methods, the segmentation results are shown in Fig. 4. Our methodology surpasses the examined approaches in solving a range of hard situations, and the outputs from our model closely match the ground truth maps, as seen in the visual comparison.

The polyps in the first two rows are markedly small. Our method shows exceptional sensitivity and precision in precisely delineating

Table 7

Ablation study of different key components on the CVC-ColonDB and Kvasir dataset.

Methods	CVC-ColonDB						Kvasir					
	$mDice$	$mIoU$	F_{β}^w	S_a	$E_{\phi}^{\max}$	$\mathcal{M}$	$mDice$	$mIoU$	F_{β}^w	S_a	$E_{\phi}^{\max}$	$\mathcal{M}$
w/o HAMB	0.728	0.658	0.712	0.827	0.877	0.035	0.893	0.839	0.878	0.909	0.943	0.029
w/o CFCM	0.747	0.673	0.729	0.833	0.875	0.036	0.893	0.838	0.880	0.907	0.937	0.033
w/o BCM	0.732	0.660	0.709	0.829	0.865	0.041	0.905	0.855	0.893	0.919	0.942	0.027
VMMCA-Net (Ours)	0.766	0.694	0.743	0.847	0.879	0.034	0.922	0.874	0.910	0.929	0.963	0.021

**Fig. 4.** The study addresses various challenging scenarios, including (a) small polyps, (b) large polyps, (c) unclear boundaries, and (d) the presence of multiple polyps.

these polyps, even if they are tiny. On the other hand, UNet and UNet++ are not able to properly segment these tiny polyps, perhaps because they cannot capture the fine-grained characteristics that are necessary for recognizing such delicate structures. The total segmentation quality is further compromised by SFA and MSNet's over-segmentation, which introduces several errors. The third and fourth rows present polyps with varying shapes and substantial sizes, which inherently complicate the segmentation task. Under these conditions, SFA and UNet++ show inferior performance compared to our method, struggling to maintain consistency across diverse polyp morphologies. Our model, leveraging advanced multi-level context aggregation and boundary-aware feature extraction, effectively handles these variations, ensuring accurate and reliable segmentation regardless of the polyps' geometric diversity. The fifth and sixth rows include visually merged polyps since there is little difference between them and the backdrop. This low-contrast scenario poses significant challenges for segmentation algorithms, often leading to misidentifications and blurred boundaries. Our approach excels in these conditions by enhancing feature differentiation and leveraging contextual information, thereby maintaining clear and precise polyp boundaries even when they blend with the background. The seventh and eighth rows contain images with multiple polyps, adding another layer of complexity to the segmentation task. Accurately segmenting multiple instances requires the model to distinguish between closely situated or overlapping polyps. Our model effectively manages this by employing robust instance differentiation mechanisms, ensuring each polyp is individually and accurately segmented without confusion or overlap. In contrast, competing methods

tend to either miss some polyps or incorrectly merge adjacent polyps, resulting in incomplete or inaccurate segmentation.

All things considered, the results of the comparison show that our strategy accomplishes better and more thorough segmentation. Specifically, our approach delineates polyps with greater precision and consistency compared to the alternative methods evaluated. The visual results confirm that our model can effectively perform polyp segmentation across a variety of challenging scenarios, including small and numerous polyps, diverse shapes and sizes, and low-contrast environments. The impressive resilience of our model is demonstrated by its ability to tackle the complex problems that arise in polyp segmentation tasks.

Our analysis also includes a comparison of the suggested model to several medical segmentation approaches that utilize Mamba. Our VMMCA-Net is compared to current visual Mamba-based algorithms in Table 6 using three benchmark datasets: Kvasir, CVC-300, and CVC-ColonDB. Our VMMCA-Net outperforms VM-UNet, VM-UNetV2, and RM-UNet on the Kvasir dataset, with $mDice$ and $mIoU$ scores of 0.922 and 0.874, respectively. The efficacy of our boundary-aware feature extraction and multi-level context aggregation in precisely defining polyp structures is demonstrated by this notable improvement. Similarly, VMMCA-Net tops the CVC-300 dataset with 0.902 on $mDice$ and 0.841 on $mIoU$. Thanks to its improved feature extraction and cross-level fusion capabilities, our model is able to handle varied polyp sizes and shapes with ease, as demonstrated by its higher performance on CVC-300. When it comes to the CVC-ColonDB dataset, VMMCA-Net has a better $mIoU$ score than RM-UNet, although RM-UNet does somewhat

Table 8

Comparison of model size and inference time.

Models	UNet [8]	UNet++ [10]	PraNet [12]	MSNet [23]	CFA-Net [31]	VMMCA-Net (ours)
Speed (FPS)	123.11	82.51	25.05	31.08	23.50	30.26
FLOPs	123.87	262.16	13.15	17.00	55.36	60.27
Param (M)	34.52	36.63	30.50	27.69	25.24	29.48

better in mDice. Despite the little discrepancy in mDice, our model still manages to keep a superior overall overlap between the ground truth and anticipated segmentation. On top of that, VMMCA-Net has better generalizability across datasets, since it routinely beats VM-UNet and VM-UNetV2 on CVC-ColonDB in every parameter.

4.3.4. Model complexity and inference time comparison

To further clarify the computational efficiency of our design, we present a comparative analysis of model complexity and inference speed in Table 8. Parameters are measured in millions (M), FLOPs in gigaflops (G), and inference speed in frames per second (FPS). Although integrating both Mamba-based global modeling and convolutional refinement, VMMCA-Net maintains a favorable efficiency–accuracy trade-off. It achieves higher throughput (30.26 FPS) than CFA-Net (23.50 FPS), while using comparable parameter counts. Although the FLOPs slightly exceed those of purely convolutional models such as UNet and UNet++, the improved structural accuracy justifies this overhead. Overall, the results demonstrate that our hybrid design enhances segmentation accuracy while incurring only a marginal computational overhead.

4.4. Ablation study

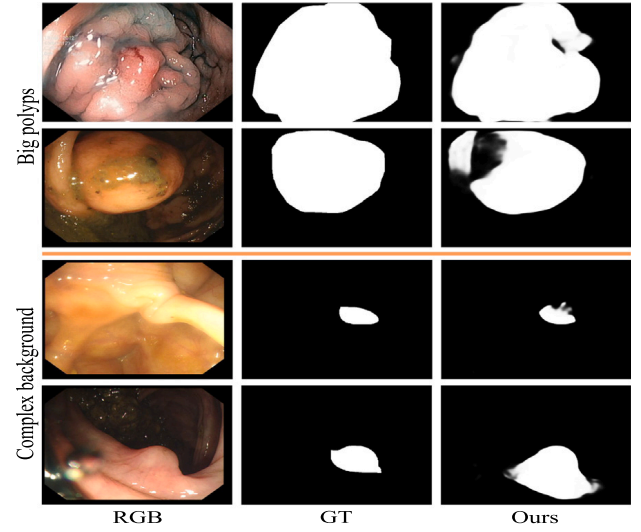
Our proposed model leverages several key modules, including the HAMB, CFCM, and BCM, to enhance segmentation performance by capturing multi-scale spatial features, cross-domain interactions, and boundary-specific information.

Effectiveness of HAMB: To evaluate the effectiveness of the HAMB, as shown in Table 7, we conduct ablation studies by removing it from the model, denoted as “w/o HAMB.” The results demonstrate that the absence of HAMB significantly reduced segmentation accuracy, indicating its crucial role in capturing multi-scale spatial features through convolutions of varying sizes, channel partitioning, and residual connections, thereby improving spatial resolution and dependency learning.

Effectiveness of CFCM: The CFCM is responsible for efficiently integrating features from different domains using multi-scale convolutions and multi-dilation encoding layers. As shown in Table 7, we assess its effectiveness by replacing the CFCM with a simple concatenation operation, referred to as “w/o CFCM.” Experimental results show that without CFCM, the model’s ability to capture cross-domain interactions and adapt to scale variations was impaired, leading to decreased segmentation precision. The CFCM module thus plays a pivotal role in facilitating the integration of cross-level features and enhancing contextual understanding.

Effectiveness of BCM: The BCM module incorporates multi-scale contextual attention and boundary-specific feature aggregation to refine the segmentation output. To verify its importance, as shown in Table 7, we conduct ablation experiments by removing BCM from the model, termed “w/o BCM.” The results indicate that BCM effectively enhances segmentation by incorporating crucial boundary information, particularly in challenging regions with complex structures. The visual comparisons further demonstrated that without BCM, boundary delineation was less precise, underscoring the importance of boundary-aware refinement for high-quality segmentation.

Overall, the combined use of HAMB, CFCM, and BCM creates a robust architecture that integrates hierarchical feature extraction, cross-domain fusion, and boundary refinement, significantly improving segmentation accuracy. Ablation studies and visual analyses confirm that

**Fig. 5.** Some failure cases obtained by the proposed VMMCA-Net.

each module contributes uniquely to the network’s ability to effectively segment complex structures, making the proposed approach well-suited for intricate segmentation tasks.

4.5. Discussion of failure cases

While qualitative and quantitative evaluations confirm the effectiveness and superiority of our VMMCA-Net, the model encounters difficulties in certain challenging scenarios, such as complex backgrounds and relatively large polyps. As illustrated in Fig. 5, the model may produce incomplete boundaries for extremely large polyps, where dominant foreground regions suppress local contrast and impede detailed boundary preservation. Furthermore, the 3rd and 4th rows demonstrate failure cases involving polyps with very low contrast that blend into the surrounding mucosa. Although our feature aggregation enhances regional consistency, the absence of clear intensity cues can cause these subtle boundaries to be missed. These observations highlight adaptive boundary enhancement for low-contrast scenarios as a key area for improvement. Consequently, addressing large-scale shape variation and robust segmentation under complex backgrounds will be a focus of our future work.

5. Conclusion

In this paper, we have proposed VMMCA-Net, a novel framework for polyp segmentation. While prior CNN- and Transformer-based methods often struggle with inconsistent cross-scale feature fusion and boundary blurring, our approach offers a more adaptive mechanism for both global context aggregation and precise edge delineation. Extensive experiments on multiple public benchmarks confirm that VMMCA-Net achieves superior performance compared to recent state-of-the-art methods. Despite its promising results, the current framework has certain limitations. The integration of multiple interaction modules increases computational complexity relative to simpler encoder-only architectures. Furthermore, while the model generalizes well across

colonoscopy datasets, its applicability to other medical imaging modalities (e.g., ultrasound or dermoscopy) requires further validation. Future work will aim to optimize computational efficiency and extend evaluation to broader clinical domains.

CRedit authorship contribution statement

Jiaye Chen: Writing – original draft, Writing – review & editing, Validation, Software, Methodology, Formal analysis. **Jian Ding:** Methodology, Investigation, Validation, Formal analysis. **Tianyang Xu:** Writing – review & editing, Conceptualization, Validation. **Rui Wang:** Visualization, Writing – review & editing Validation. **Xiaoning Song:** Supervision, Writing – review & editing, Project administration, Conceptualization. **Tao Zhou:** Writing – review & editing, Supervision, Resources, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

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Data availability

Data will be made available on request.

References

- [1] M. Navarro, A. Nicolas, A. Ferrandez, A. Lanás, Colorectal cancer population screening programs worldwide in 2016: An update, *World J. Gastroenterol.* 23 (20) (2017) 3632.
- [2] X. Guo, C. Yang, Y. Liu, Y. Yuan, Learn to threshold: Thresholdnet with confidence-guided manifold mixup for polyp segmentation, *IEEE Trans. Med. Imaging* 40 (4) (2020) 1134–1146.
- [3] V. Sharma, A. Kumar, D. Jha, M.K. Bhuyan, P.K. Das, U. Bagci, Controlpolypnet: Towards controlled colon polyp synthesis for improved polyp segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 2325–2334.
- [4] Z. Lu, Y. Zhang, Y. Zhou, Y. Wu, T. Zhou, Domain-interactive contrastive learning and prototype-guided self-training for cross-domain polyp segmentation, *IEEE Trans. Med. Imaging* 44 (9) (2025) 3563–3573.
- [5] J. Mei, T. Zhou, K. Huang, Y. Zhang, Y. Zhou, Y. Wu, H. Fu, A survey on deep learning for polyp segmentation: Techniques, challenges and future trends, *Vis. Intell.* 3 (1) (2025) 1.
- [6] M. Akbari, M. Mohrekeh, E. Nasr-Esfahani, S.R. Soroushmehr, N. Karimi, S. Samavi, K. Najarian, Polyp segmentation in colonoscopy images using fully convolutional network, in: *International Conference of the IEEE Engineering in Medicine and Biology Society*, IEEE, 2018, pp. 69–72.
- [7] X. Sun, P. Zhang, D. Wang, Y. Cao, B. Liu, Colorectal polyp segmentation by U-net with dilation convolution, in: *IEEE International Conference on Machine Learning and Applications*, IEEE, 2019, pp. 851–858.
- [8] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2015, pp. 234–241.
- [9] D. Jha, P.H. Smedsrud, M.A. Riegler, D. Johansen, T. De Lange, P. Halvorsen, H.D. Johansen, Resunet++: An advanced architecture for medical image segmentation, in: *IEEE International Symposium on Multimedia*, IEEE, 2019, pp. 225–2255.
- [10] Z. Zhou, M.M.R. Siddiquee, N. Tajbakhsh, J. Liang, Unet++: Redesigning skip connections to exploit multiscale features in image segmentation, *IEEE Trans. Med. Imaging* 39 (6) (2019) 1856–1867.
- [11] Y. Fang, C. Chen, Y. Yuan, K.-y. Tong, Selective feature aggregation network with area-boundary constraints for polyp segmentation, in: *International Conference on Medical Image Computing and Computer Assisted Intervention*, Springer, 2019, pp. 302–310.
- [12] D.-P. Fan, G.-P. Ji, T. Zhou, G. Chen, H. Fu, J. Shen, L. Shao, Pranet: Parallel reverse attention network for polyp segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2020, pp. 263–273.
- [13] B. Kayalibay, G. Jensen, P. Van Der Smagt, CNN-based segmentation of medical imaging data, 2017, arXiv preprint arXiv:1701.03056.
- [14] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H.R. Roth, D. Xu, Unetr: Transformers for 3d medical image segmentation, in: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 574–584.
- [15] H.-Y. Zhou, J. Guo, Y. Zhang, X. Han, L. Yu, L. Wang, Y. Yu, Nnformer: Volumetric medical image segmentation via a 3d transformer, *IEEE Trans. Image Process.* 32 (2023) 4036–4045.
- [16] G. Litjens, T. Kooi, B.E. Bejnordi, A.A.A. Setio, F. Ciompi, M. Ghafoorian, J.A. Van Der Laak, B. Van Ginneken, C.I. Sánchez, A survey on deep learning in medical image analysis, *Med. Image Anal.* 42 (2017) 60–88.
- [17] D. Zhang, G. Huang, Q. Zhang, J. Han, J. Han, Y. Yu, Cross-modality deep feature learning for brain tumor segmentation, *Pattern Recognit.* 110 (2021) 107562.
- [18] A. Oulefki, S. Agaian, T. Trongtirakul, A.K. Laouar, Automatic COVID-19 lung infected region segmentation and measurement using CT-scans images, *Pattern Recognit.* 114 (2021) 107747.
- [19] N. Tajbakhsh, S.R. Gurudu, J. Liang, Automated polyp detection in colonoscopy videos using shape and context information, *IEEE Trans. Med. Imaging* 35 (2) (2015) 630–644.
- [20] Y. Pang, X. Zhao, L. Zhang, H. Lu, Multi-scale interactive network for salient object detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 9413–9422.
- [21] M.M. Rahman, M. Munir, R. Marculescu, Emcad: Efficient multi-scale convolutional attention decoding for medical image segmentation, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 11769–11779.
- [22] K. Jiang, Z. Wang, P. Yi, C. Chen, B. Huang, Y. Luo, J. Ma, J. Jiang, Multi-scale progressive fusion network for single image deraining, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 8346–8355.
- [23] X. Zhao, L. Zhang, H. Lu, Automatic polyp segmentation via multi-scale subtraction network, in: *International Conference on Medical Image Computing and Computer Assisted Intervention*, Springer, 2021, pp. 120–130.
- [24] G. Yue, G. Zhuo, W. Yan, T. Zhou, C. Tang, P. Yang, T. Wang, Boundary uncertainty aware network for automated polyp segmentation, *Neural Netw.* 170 (2024) 390–404.
- [25] T.-C. Nguyen, T.-P. Nguyen, G.-H. Diep, A.-H. Tran-Dinh, T.V. Nguyen, M.-T. Tran, Cebanet: cascading context and balancing attention for polyp segmentation, in: *International Conference on Medical Image Computing and Computer Assisted Intervention*, Springer, 2021, pp. 633–643.
- [26] J. Li, J. Wang, F. Lin, A.A. Heidari, Y. Chen, H. Chen, W. Wu, Prcnet: A parallel reverse convolutional attention network for colorectal polyp segmentation, *Biomed. Signal Process. Control.* 95 (2024) 106336.
- [27] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: *Proceedings of the European Conference on Computer Vision*, 2018, pp. 801–818.
- [28] M. Tan, R. Pang, Q.V. Le, Efficientdet: Scalable and efficient object detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 10781–10790.
- [29] S. Zheng, J. Lu, H. Zhao, X. Zhu, Z. Luo, Y. Wang, Y. Fu, J. Feng, T. Xiang, P.H. Torr, et al., Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 6881–6890.
- [30] A. Sinha, J. Dolz, Multi-scale self-guided attention for medical image segmentation, *IEEE J. Biomed. Health Informatics* 25 (1) (2020) 121–130.
- [31] T. Zhou, Y. Zhou, K. He, C. Gong, J. Yang, H. Fu, D. Shen, Cross-level feature aggregation network for polyp segmentation, *Pattern Recognit.* 140 (2023) 109555.
- [32] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, P. Torr, Res2net: A new multi-scale backbone architecture, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (2) (2019) 652–662.
- [33] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, X. Wang, Vision mamba: Efficient visual representation learning with bidirectional state space model, 2024, arXiv preprint arXiv:2401.09417.
- [34] J. Ruan, S. Xiang, Vm-unet: Vision mamba unet for medical image segmentation, 2024, arXiv preprint arXiv:2402.02491.
- [35] D. Ouyang, S. He, G. Zhang, M. Luo, H. Guo, J. Zhan, Z. Huang, Efficient multi-scale attention module with cross-spatial learning, in: *IEEE International Conference on Acoustics, Speech and Signal Processing*, IEEE, 2023, pp. 1–5.
- [36] E. Xie, W. Wang, W. Wang, M. Ding, C. Shen, P. Luo, Segmenting transparent objects in the wild, in: *European Conference on Computer Vision*, Springer, 2020, pp. 696–711.

- [37] J. Bernal, F.J. Sánchez, G. Fernández-Esparrach, D. Gil, C. Rodríguez, F. Vilariño, WM-DOVA maps for accurate polyp highlighting in colonoscopy: Validation vs. saliency maps from physicians, *Comput. Med. Imaging Graph.* 43 (2015) 99–111.
- [38] J. Silva, A. Histace, O. Romain, X. Dray, B. Granado, Toward embedded detection of polyps in wce images for early diagnosis of colorectal cancer, *Int. J. Comput. Assist. Radiol. Surg.* 9 (2014) 283–293.
- [39] D. Jha, P.H. Smedsrud, M.A. Riegler, P. Halvorsen, T. De Lange, D. Johansen, H.D. Johansen, Kvasir-seg: A segmented polyp dataset, in: *International Conference on MultiMedia Modeling*, Springer, 2020, pp. 451–462.
- [40] D. Vázquez, J. Bernal, F.J. Sánchez, G. Fernández-Esparrach, A.M. López, A. Romero, M. Drozdal, A. Courville, A benchmark for endoluminal scene segmentation of colonoscopy images, *J. Heal. Eng.* 2017 (1) (2017) 4037190.
- [41] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, A. Borji, Enhanced-alignment measure for binary foreground map evaluation, 2018, *arXiv preprint arXiv:1805.10421*.
- [42] R. Zhang, G. Li, Z. Li, S. Cui, D. Qian, Y. Yu, Adaptive context selection for polyp segmentation, in: *International Conference on Medical Image Computing and Computer Assisted Intervention*, Springer, 2020, pp. 253–262.
- [43] Y. Sun, G. Chen, T. Zhou, Y. Zhang, N. Liu, Context-aware cross-level fusion network for camouflaged object detection, 2021, *arXiv preprint arXiv:2105.12555*.
- [44] R. Zhang, P. Lai, X. Wan, D.-J. Fan, F. Gao, X.-J. Wu, G. Li, Lesion-aware dynamic kernel for polyp segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2022, pp. 99–109.
- [45] G. Yue, H. Xiao, T. Zhou, S. Tan, Y. Liu, W. Yan, Progressive feature enhancement network for automated colorectal polyp segmentation, *IEEE Trans. Autom. Sci. Eng.* 22 (2024) 5792–5803.
- [46] Z. Yu, L. Zhao, T. Liao, X. Zhang, G. Chen, G. Xiao, A novel non-pretrained deep supervision network for polyp segmentation, *Pattern Recognit.* 154 (2024) 110554.
- [47] M. Zhang, Y. Yu, S. Jin, L. Gu, T. Ling, X. Tao, VM-UNET-V2: rethinking vision mamba unet for medical image segmentation, in: *International Symposium on Bioinformatics Research and Applications*, Springer, 2024, pp. 335–346.
- [48] H. Tang, G. Huang, L. Cheng, X. Yuan, Q. Tao, X. Chen, G. Zhong, X. Yang, Rm-unet: Unet-like mamba with rotational ssm module for medical image segmentation. *Signal, Image Video Process.* (2024) 1–17.